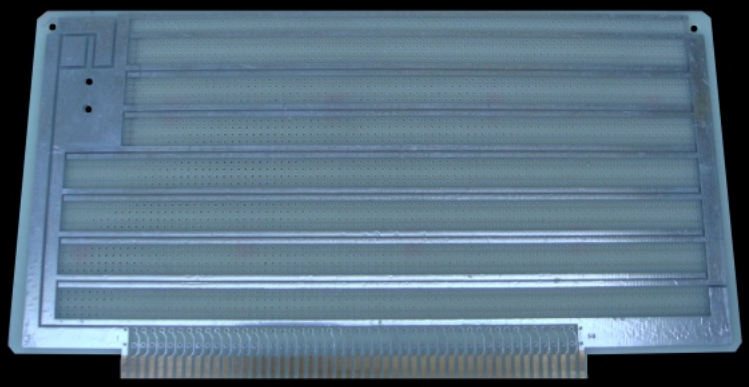
An S-100 Prototype Board

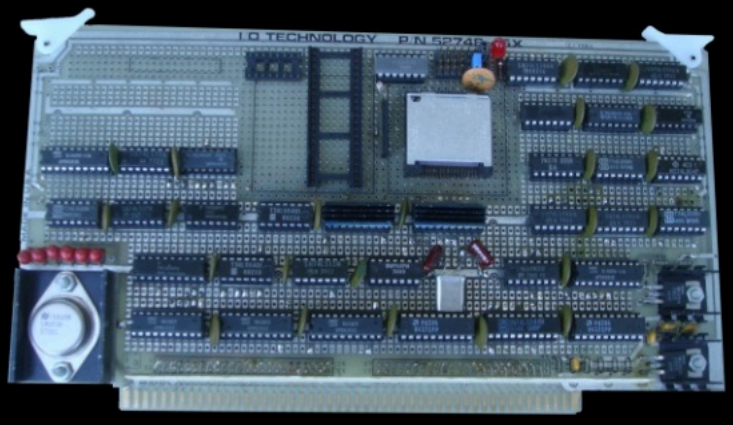
Introduction

If you do any kind of serious "homebrew" work with your S-100 system you will need a "Prototype Board". This is a card that fits into your S-100 bus and brings the bus edge connector pins (all 100 of them) up on to the board. Almost from the very start companies supplied these boards for what was at the time a hobbyists market. Here is a picture of a typical board:-



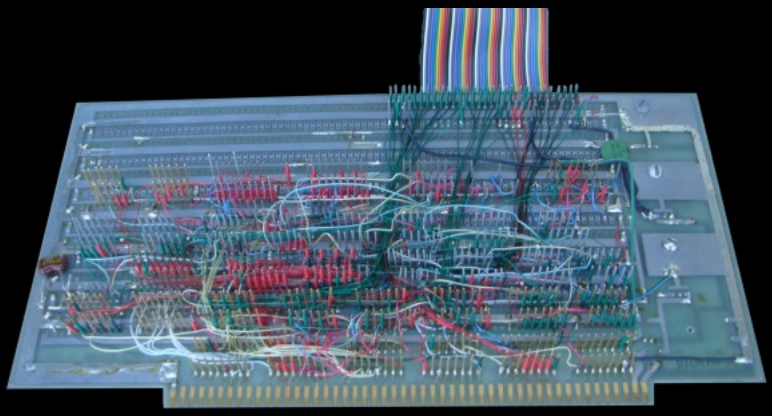
Late 70's Era S-100 Bus Prototype Board

Typically you laid out the chips horizontally (although some vertical orientation boards were produced). These board can be found on places like eBay from time to time. When later non-DIP style chips (CPU's etc.) started to appear these boards did not work. One or two companies offered boards with just 0.1" pad spacing or a board with at least an area with this format such as this I/O Technology board.

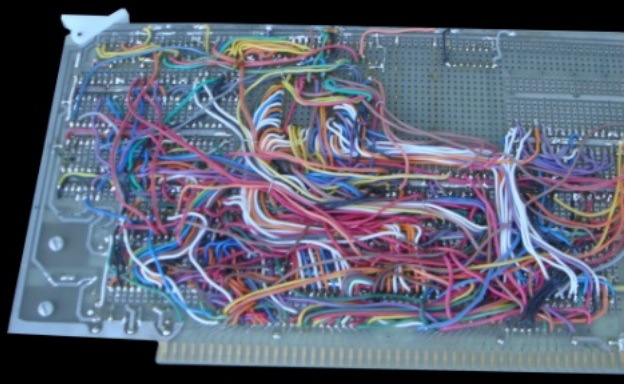


I/O Technology Prototype Board

Connecting the IC pins together can be done either via *wire wrap* or *point to point solder* techniques. There are pros and cons to each method. Here are two examples:-



Wire Wrap

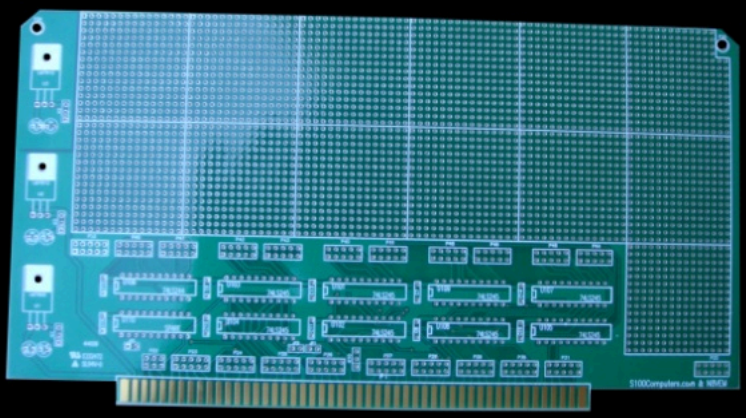


Point to Point Solder Connections

Personally I prefer the point to point soldering method. It takes longer to do initially but in the end you save time debugging because it is easier to locate connections and make changes.

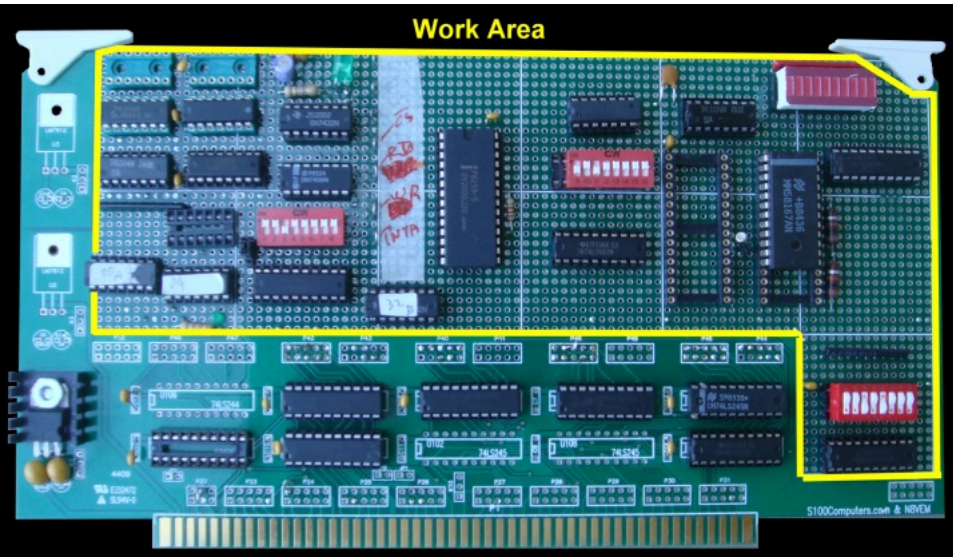
A Buffered S-100 Prototype Board

One of the frustrating aspects of doing a new prototype board design is that before you can do anything useful you have to connect up all the address lines (24) , data lines (16) and numerous control line to buffers on the board. Each requires an input line from the bus and an output line to the functional area of the board. For almost every board they will be the same. To take the drudgery out of this a "buffered prototype S-100" board with all these connections already done. Here is a picture of the board:-



S100Computers.com & N8VEM Prototype Board

There are distinct pads for all the major buffered S-100 lines. You can start right away utilizing the board for what you had in mind. The schematic for the board is quite simple and can be seen here. The layout for the board can be seen here. There are pads for a +5V, +12V and -5V or -12V voltage regulator. The only slightly unusual thing is U108, a 74LS245 buffer that may be used to transfer data to/from the 8 bit data In/Out lines. This can sometimes be of use for interrupts in 16 bit systems. Unfortunately its on the S-100 side of bus buffers and so cannot be used for 16 bit CPU data transfers if you wanted to use this board as an 16 Bit CPU prototype board. I have use this board for a number of S-100 prototype boards. Here is a picture of it being used for the first PIC/RTC board:-

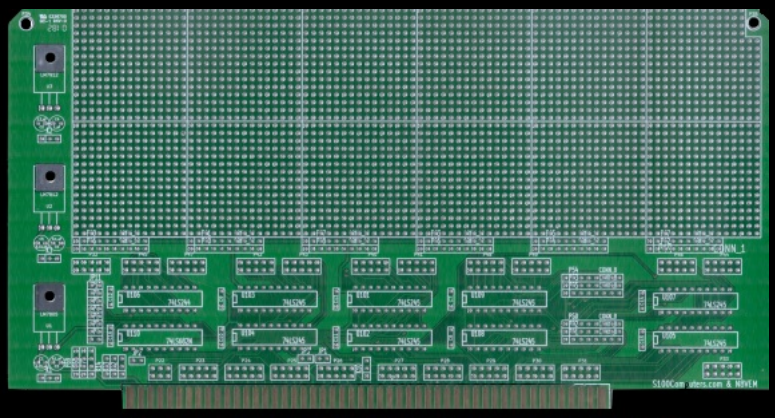


S-100 Prototype Board

V2 of the Buffered S-100 Prototype Board.

Having utilized a number of these boards, I realized that a few little tweaks would make a big difference. A second version of the board was made. This board has in addition to the board above:-

1. I/O port selection circuitry that can be jumpered for any 8 bit I/O address range.
2. A complete bi-directional buffer (U108) setup for 8/16 bit data transfers with 16 bit CPU's.
3. A dedicated 5V line across the bottom of the "patch" area for convenient hookup to IC's.
4. Complete flexibility as to the type or Voltage regulators used in each of the 3 positions.



S100Computers & N8VEM S-100 Prototype Board V2

The schematic for the V2 Prototype board can be obtained here. The board layout can be obtained here.
The board utilizes 74LS682's for port addressing. If your are unfamiliar with this technique click here.

A Production S-100 Board

Realizing that a number of people might want to utilize a board like this together with a group of people on the Google Groups S100Computers Forum, "group purchases" are made from time to time.

Please see here for more information.

The links below will contain the most recent schematic of this board.
Note, it may change over time and some IC part or pin numbers may not correlate *exactly* with the text in the article above.
MOST CURRENT PROTOTYPE BOARD SCHEMATIC (V2, FINAL, 7/4/2010)
MOST CURRENT PROTOTYPE BOARD LAYOUT (V2, FINAL, 7/4/2010)
Most current KiCAD files for this board (S100 Prototyping Board V3.0a 10/13/2017)
Most current Gerber files for this board (S100 Prototyping Board V3.0a 10/13/2017)

Other pages describing my S-100 hardware and software.
Please click here to continue...

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